

Trainings ▾

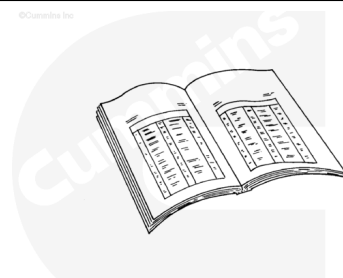
Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



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Note : The aftertreatment fuel injector fuel and coolant lines may have enough flexibility to obtain access to clean the injector tip.

Note : Determine the appropriate radiator or coolant expansion tank cap adapter for the vehicle being serviced. A list of available caps and applications can be found in Section 14 of the Service Products Catalog, Bulletin 3377710.

Use a vacuum pump type device on the cooling system. Install the adapter on the radiator or coolant expansion tank. Connect an automotive hand held vacuum pump and apply 7.26 cm-Hg [3 in-Hg] of vacuum.

Note : Depending on the cooling system configuration, more vacuum may be required to keep the cooling system in balance.

Place a container under the aftertreatment fuel injector. There may be a small quantity of coolant that drains from the coolant lines when disconnected.

- If cleaning the injector off-engine, drain the coolant.
- Disconnect the fuel lines and coolant lines **only** if removing the injector from the engine for cleaning or replacement.

- Disconnect the wiring harness from the aftertreatment injector.

Remove

Note : The aftertreatment fuel injector fuel and coolant lines may have enough flexibility to obtain access to clean the injector tip without fully removing the aftertreatment fuel injector from the engine.

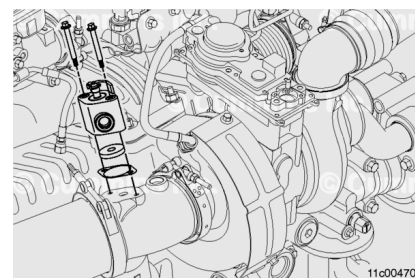
Remove and discard the capscrews holding the injector.

Remove the injector assembly.

Discard the metal gasket.

Discard the capscrews.

Discard the fibrous insulator.



Clean

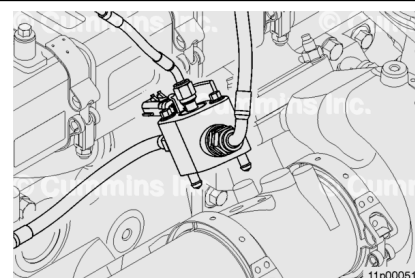
On-Engine

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : The aftertreatment fuel injector fuel and coolant lines may have enough flexibility to obtain access to clean the injector tip without fully removing the aftertreatment fuel injector from the engine.

Loosen the two mounting capscrews and lift the injector to obtain access to the injector tip.

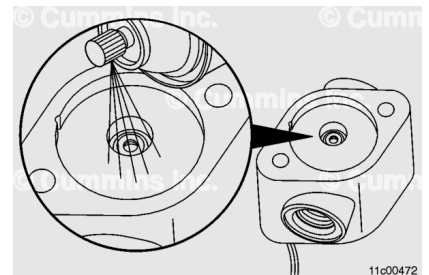
Discard the metal gasket, fibrous isolator, and capscrews.

Cover the aftertreatment fuel injector mounting surface on the aftertreatment adapter pipe with tape to avoid allowing debris to enter the exhaust system.

Spray the tip of the aftertreatment fuel injector with carburetor cleaner. When spraying, focus the spray on the tip of the injector.

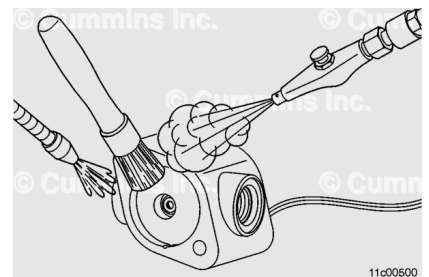
Use a brass brush to brush the tip of the injector for 15 seconds. Repeat spraying and brushing three times. Often the carbon will be removed from the tip in a short period, but repeating the cleaning will remove the carbon built up in the nozzle.

Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.



⚠ CAUTION ⚠

Do not use a nylon brush. Use only a stiff brass brush with undamaged bristles to clean the tip of the aftertreatment fuel injector. The use of a brush with damaged bristles will not be as effective. The use of a steel brush or a steel wheel will cause permanent damage to the aftertreatment fuel injector.



Use a brass brush to agitate the carburetor cleaner, focusing on the tip of the injector.

Repeat spraying and brushing as necessary to remove any carbon from the tip of the aftertreatment fuel injector.

Note : The carbon this procedure is attempting to remove can often **not** be seen with the unaided eye. This carbon builds up in the nozzle tip area of the injector.

Off-Engine

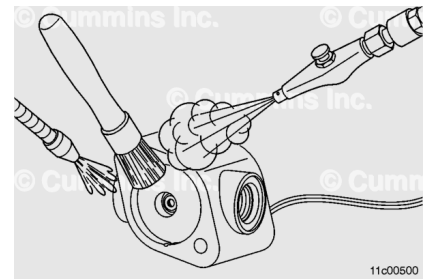
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



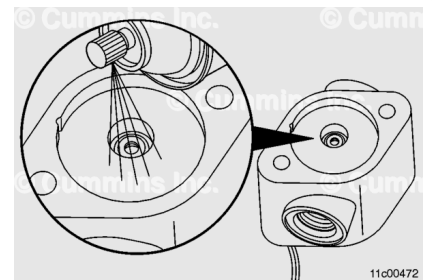
Inspect the injector tip.

Some carboning can occur on the tip of the injector.

Spray the tip of the aftertreatment fuel injector with carburetor cleaner. When spraying, focus the spray on the tip of the injector. Use enough carburetor cleaner to fill the cavity surrounding the tip and to cover the tip.

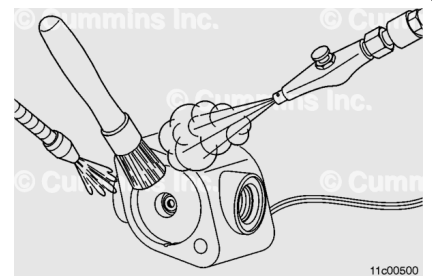
Allow the carburetor cleaner to sit for 15 seconds so that it can break down and penetrate the debris that has collected in the tip area.

Note : Carburetor cleaner is the most effective solvent for this procedure. Do not substitute any other cleaning solvent.



⚠ CAUTION ⚠

Do not use a nylon brush. Use only a stiff brass brush with undamaged bristles to clean the tip of the aftertreatment fuel injector. The use of a brush with damaged bristles will not be as effective. The use of a steel brush or a steel wheel will cause permanent damage to the aftertreatment fuel injector.



Use a brass brush to agitate the carburetor cleaner, focusing on the tip of the injector.

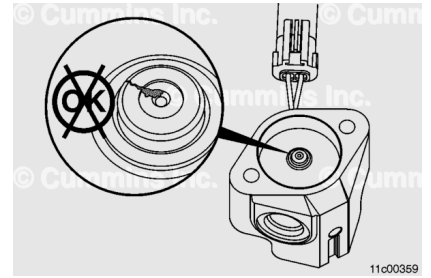
Repeat spraying and brushing for 1 minute. Often the carbon will be removed from the tip in a short period, but cleaning for a full minute will remove the carbon built up in the nozzle.

Note : The carbon this procedure is attempting to remove can often not be seen with the unaided eye. This carbon builds up in the nozzle tip area of the injector.

Inspect for Reuse

Inspect the injector tip for cracks or other damage. Replace the injector, if necessary.

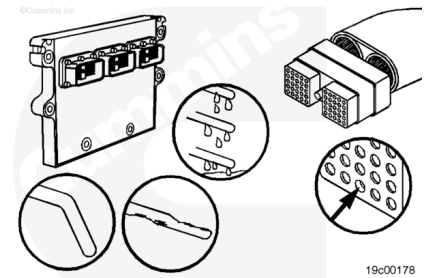
Clean and inspect the adapter tube injector mounting surface.



If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to a Cummins® Authorized Repair Location.

Inspect the harness and connector pins for the following:

- Loose connector
- Corroded pins
- Bent or broken pins
- Pushed back or expanded pins
- Moisture in or on the connector
- Missing or damaged connector seals
- Dirt or debris in or on the connector pins
- Connector shell broken
- Wire insulation damage
- Damaged connector locking tab.



Use the following procedure for general inspection techniques.

Refer to Procedure 019-361 in Section 19.

(/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)

Install

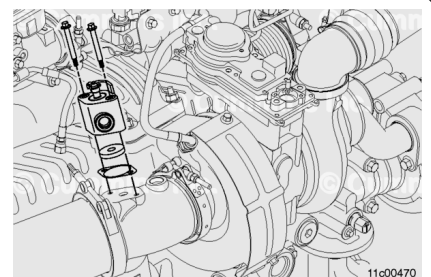
Install a new fibrous insulator.

Install a new metal gasket.

Note : The fibrous insulator will be against the injector.

Install the aftertreatment injector onto the adapter pipe.

Note : Make sure the aftertreatment injector harness is not touching the adapter pipe.



Note : The tall side of the injector and injector heat shield should be facing the turbocharger side.

Install new capscrews. Anti-seize compound will already be applied to the capscrews. The use of the original capscrews will result in clamp load loss and gasket leakage.

Torque Value: 11.3 n•m [100 in-lb]

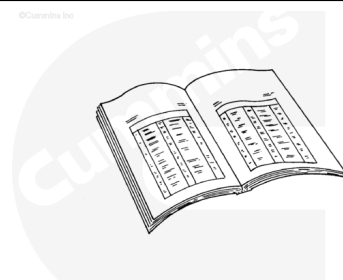
Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Connect the fuel line and coolant lines.
- If disconnected, connect the wiring harness to the aftertreatment injector.
- If the coolant was drained, fill the cooling system.
- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to a Cummins® Authorized Repair Location.



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